the gap between the days it rains is also increasing. Between July 1 and September 19, for instance, there were consecutive dry days for two to three weeks in Rajasthan, Gujarat, Maharashtra, Karnataka, Andhra Pradesh, Tamil Nadu, Punjab and Haryana. A dry day is one with less than 2.5 mm of rainfall in a 25-km grid.

The outlook could get worse. The World Meteorological Organization (WMO) has forecast El Niño to strengthen to very strong intensity towards the end of this year, raising the risk of floods, drought and extreme heat through 2027. The fallout could extend to food security. External affairs minister S. Jaishankar, speaking at an event in New York on September 28, warned that the world faces a "major food crisis" in the coming months, as wars in Ukraine and the Middle East compound the effects of a super El Niño.

AGRARIAN IMPACT

The changing pattern of rainfall isn't optimal for crops and plants. According to Karthikeyan Lanka, associate professor at IIT Bombay's Centre for Climate Studies, soil has a water-holding capacity; anything beyond that just washes off. "Extreme rainfall followed by dry spells does not provide optimal conditions for plants to survive," he says, "because they are getting too much rain at one time but are dry for the rest of the days, just like how you don't put an entire bucket of water in your garden plant in one go. If you do, it dies."

A 2024 study published in the *Communications Earth & Environment* journal that looked at Indian rice production and climate data from 1990-2017 found that rainfall deficits reduced rice yields, but excessive rainfall caused even larger losses, decreasing rice yield by 33.7 per cent, compared to deficit rainfall, which impacted yield by 19 per cent.

Beyond water scarcity for agriculture in the current season, drought can affect the next 2-3 agricultural cycles as well. That is what makes it particularly dangerous: its effects spread across agriculture, water availability, groundwater recharge, energy demand as hydroelectricity is affected, and irrigation, says Mishra.



"You may see a lot more drought and floods in the same season, partly because of climate change"

VIMAL MISHRA
Professor & Dean, R&D, IIT Gandhinagar



"It is not just the lack of rainfall that is contributing to drought, anthropogenic reasons, too, are increasing its impact. That's the double whammy that we are talking about"

ABINASH MOHANTY
Global Head, Climate Change and Sustainability, IPE Global

And it is not just the lack of rainfall that is contributing to drought, anthropogenic reasons, too, are increasing its impact. Excessive urbanisation has slowed groundwater percolation, while extensive construction and infrastructure development are leading to more use of groundwater, which isn't getting recharged. As a result, major urban centres like Bengaluru, Mumbai and Chennai, which all get adequate rainfall, have been facing a water crisis. "That's the double whammy or the double conundrum that we are talking about," says Abinash Mohanty, global head, climate change and sustainability, at development consultancy IPE Global.

REACTIVE TO PROACTIVE

Unlike other extreme weather events, drought doesn't happen suddenly but manifests over weeks and months. Yet India's policy on managing drought remains essentially reactive, with relief to farmers for crop damage coming after its onset is declared.

The drought code, says G.V. Ramanjaneyulu, executive director of the Hyderabad-based Centre for Sustainable Agriculture, is written from a famine point of view, wherein once the state announces a drought, the Centre verifies the damage through a ground visit and announces the compensation. "This process is like a post-mortem as it takes 1-2 years before the farmer receives the relief," he says.

Since drought is an annual occurrence in some or the other part of the country, there is a need to move from a reactive strategy to adaptation and preparedness. Recently, agriculture minister Chouhan asked states to plan their rabi crop sowing based on available water and soil moisture in their region. Ramanjaneyulu emphasises the need for better forecasting so that advisories reach farmers before they buy seeds, and continue through the season to guide cropping patterns, production practices and insurance mechanisms.

India still lacks proactive systems for data management. Even with IT systems, drones and satellite maps making it easier to map drought, this capability has not been integrated into the government machinery. Advance information on where drought conditions are likely to persist can help decision-makers and communities protect water resources, plan crops and sustain the livelihoods. The priority, Ramanjaneyulu says, should be to translate these forecasts into information that governments and farmers can use. ■

— with bureau inputs